

Group Project Document Requirements: Final Copy

“Is healthcare more expensive for cisgender men vs women?”

INTRODUCTION AND BACKGROUND

Cisgender men and women interact with the healthcare system in distinct ways, shaped by behavioural patterns, biological differences and social factors. These differences could suggest that this leads to unequal health needs, outcomes and NHS spending. Yet, public datasets rarely offer a clear, integrated view of how healthcare costs vary by biological sex across the life course. To note, our report did not account for trans and non-binary identities as such NHS/ONS data were not available, so for argument's sake, we focused solely on the data present for male and female.

Why this problem matters (and why it's realistic)

Public NHS and ONS datasets are easily accessible, structured and well-documented. This made the project feasible within the timeline and well-suited to the skills taught in the Data Specialisation module, such as data cleaning, joining datasets, dashboarding, and storytelling.

Analysing sex-based cost differences helped to illuminate:

- Differences in healthcare usage (including use of private healthcare referrals and providers)
- Variations in risk factors (location, lifestyle, age)
- Potential inequities or unmet needs (help-seeking behavioural patterns in data)
- Areas where one sex faced a greater medical or financial burden.

Understanding these patterns would better support our target audiences in NHS resource planning and service design, through the use of simple non-technical visualisations.

Our project aimed to create a clear data-driven comparison of sex-based healthcare costs by combining multiple NHS and ONS datasets. Using Jupyter Notebook, we built a narrative dashboard that highlighted how costs evolved over time, which conditions drove them, and where usage patterns differed. By combining usage patterns, cost data, and health context, the project formed a picture of how sex-based differences in healthcare spending emerged.

Our project found that healthcare costs are not driven by price alone but also by healthcare usage patterns. Differences in how often individuals use services such as emergency departments, hospitals, etc., affect healthcare spending. Understanding whether one sex uses healthcare services more frequently, complex issues or for sex- specific issues would help to explain cost differences between the sexes.

This project also recognised that health begins before hospital treatment, so the use of population health indicators such as general well-being and long-term conditions, offered important context to interpret the demand for healthcare and outcomes.

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SPECIFICATIONS AND DESIGN

We aimed to have answered three core questions:

- *Do cisgender men or women incur higher overall NHS Healthcare costs?*
- *Which conditions, services, or age bands contribute most to these differences?*
- *What does the landscape of public and private healthcare spending look like, and how has it changed over time?*

We answered these by:

- **Comparing sexes across conditions, age groups and service categories**, focusing on how cost patterns diverge.
- **Analysing usage patterns** in key areas such as:
 - General Health
 - Hospital Admissions
 - Maternity care
 - Emergency care
 - Long-term conditions
- **Linked cost drivers to demographic and lifestyle factors**
- **Analysed expenditure by finance scheme overtime**
- **Presented findings through visually accessible means** rather than technical modelling.

The project followed a simple data analysis workflow:

- **Data was sourced primarily from NHS and ONS datasets**, ensuring reliability and consistency across sources.
- **Each dataset was individually cleaned and processed**, addressing formatting differences and preparing them for integrated analysis.
- **Findings were interpreted using charts, comparisons and narrative explanations**, enabling a clear understanding of cost and utilisation patterns by sex.
- **Analysis was completed within Jupyter notebooks**, supporting reproducibility and collaborative work.
- **Outputs were stored and version-controlled in GitHub**, enabling transparent collaboration and ensuring all team members work from aligned documents and code.

The workflow was designed to maintain:

- **Accurate and rigorous data handling** across all datasets
- **Clear and effective presentation of results**, suited to non-technical audiences
- **Strong teamwork and robust version control** to ensure smooth collaboration and traceability across the project.

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IMPLEMENTATION AND EXECUTION

The project was delivered collaboratively, with assigned key responsibilities to members of the team based on interest and experience. The team was split into three sub-groups to focus on analysing the three core datasets. This allowed each member of the team to gain experience in the full process of data cleaning, exploration, analysis and visualisation. The team maintained alignment through regular meetings and shared documentation.

Responsibility	How many team members	Who
Project definition and proposal	All team, 1 for final edit	May
Meeting coordinator and compilation of group homework	1	May
Kanban board manager	1	Selena
Contributions of sprint reviews and retrospective	1	Selena
Data cleaning	All team	3 sub-teams
Data merging and transformation	All team	3 sub-teams
Descriptive analysis and visualisation	All team	3 sub-teams
API Implementation	1	Selena
Compilation of final Jupyter Notebook	3	Heather, May, Kat
Compilation of Readme.md file	2	Sherelle, Rhea, Fola
Compilation of the Project Report	All team, 1 for final edit	May
Compilation of Presentation	3	May, Kat, Fola
GitHub management	2	Heather, May

We distributed the workload based on each member's interests and strengths, guided by the poll we completed to identify preferred areas. Tasks were assigned fairly and transparently, ensuring everyone contributed to the sections they felt most confident in.

To stay organised, we used:

- **Trello** to track tasks, deadlines and progress
- **Weekly check-ins** to review updates and align priorities
- **Polly polls** for quick team decisions
- **Google Drive and Teams** for shared documents and communication

This structure ensured a balanced workload, consistent momentum and clear accountability.

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Libraries and Languages used

- Python
- Pandas
- Numpy
- Matplotlib
- Seaborn

Version Control

How we managed our code

- A GitHub repository was created to store raw and cleaned datasets, Jupyter Notebook files for cleaning and reprocessing, as well as analysis and visualisations and supporting assets.
 - This ensured code quality, reproducibility and smooth collaboration
- We decided to split into three sub-teams to tackle the three primary datasets, and used Google Colab to allow team members to collaborate in real time.
 - The Colab notebooks were saved in our shared Google Drive, so all team members could view and access shared code.
- Once the code was completed in Colab, a member from each sub-team transposed and pushed the code to the Jupyter Notebook in the GitHub repository.

DATA COLLECTION

Primary and Secondary Dataset Source Retrieval

The project required publicly available datasets containing breakdowns by sex, age, care type and cost. These primary datasets were downloaded directly from the NHS and ONS and were stored as raw data in both the shared Google Drive and GitHub. These provided high-quality data for such purposes, though inconsistencies across datasets made careful cleaning essential.

Primary datasets made up the bulk of our analysis, while secondary datasets were used where relevant and appropriate.

Cleaning and Pre-processing

Pre-processing addressed numerous issues, such as:

- ❖ Embedded metadata
- ❖ Duplicate or subtotal rows
- ❖ Missing values
- ❖ Inconsistent formatting

To name a few, whilst some datasets required the addition of categorical columns. Differences in formatting across years meant that proportions rather than raw totals were sometimes used for comparison.

API

Hospital admissions data were retrieved via the Fingertips API. The API data was accessed through the *fingertips_py* Python package.

- [Emergency Hospital Admissions for Intentional and Unintentional Injuries \(via Fingertips\)](#)

The Fingertips API dataset required a different preparation approach due to its structure. This included:

- Retrieving data via the *fingertip_py* Python package
- Extracting key fields (time, sex, area and value)
- Removing duplicate national records
- Removing combined sex categories

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Primary Datasets:

- [General health by age, sex and deprivation, England & Wales \(Census 2021\)](#): Shows self-reported general health by sex and deprivation. Useful for SES (socio-economic status) + gender interplay.
- [Hospital Admitted Patient Care Activity, 2023-24 - NHS England Digital](#): Data by age, sex, diagnosis, procedure. Great for gender differences in hospital care utilisation.
- [Patient Level Activity and Costing, 2021-22](#): Contains cost & activity data (acute, mental health, ambulance) – can compare male vs female cost usage within range of age groups.

Secondary Datasets (links embedded): *These were supplementary to use as and when needed in aid of our arguments*

- [UK Health Accounts 2024](#) Used to provide a contextual background to healthcare spending by financing scheme, function and provider
- [Acute Patient Level Activity and Costing, 2019-20](#) Used to compare to 2021-22 data to explore spending changes year-on-year

Anticipated challenges

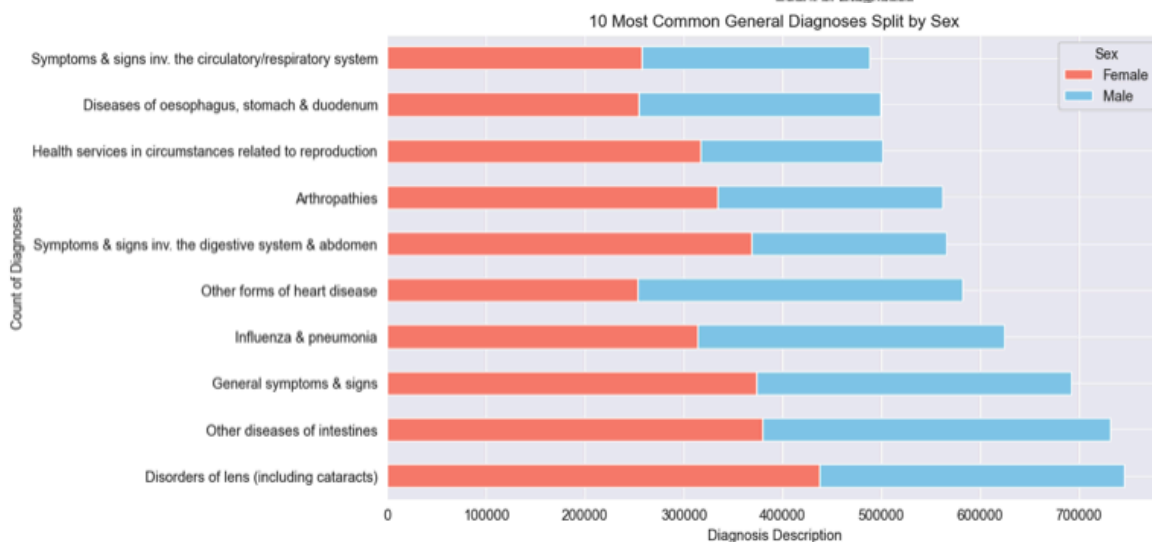
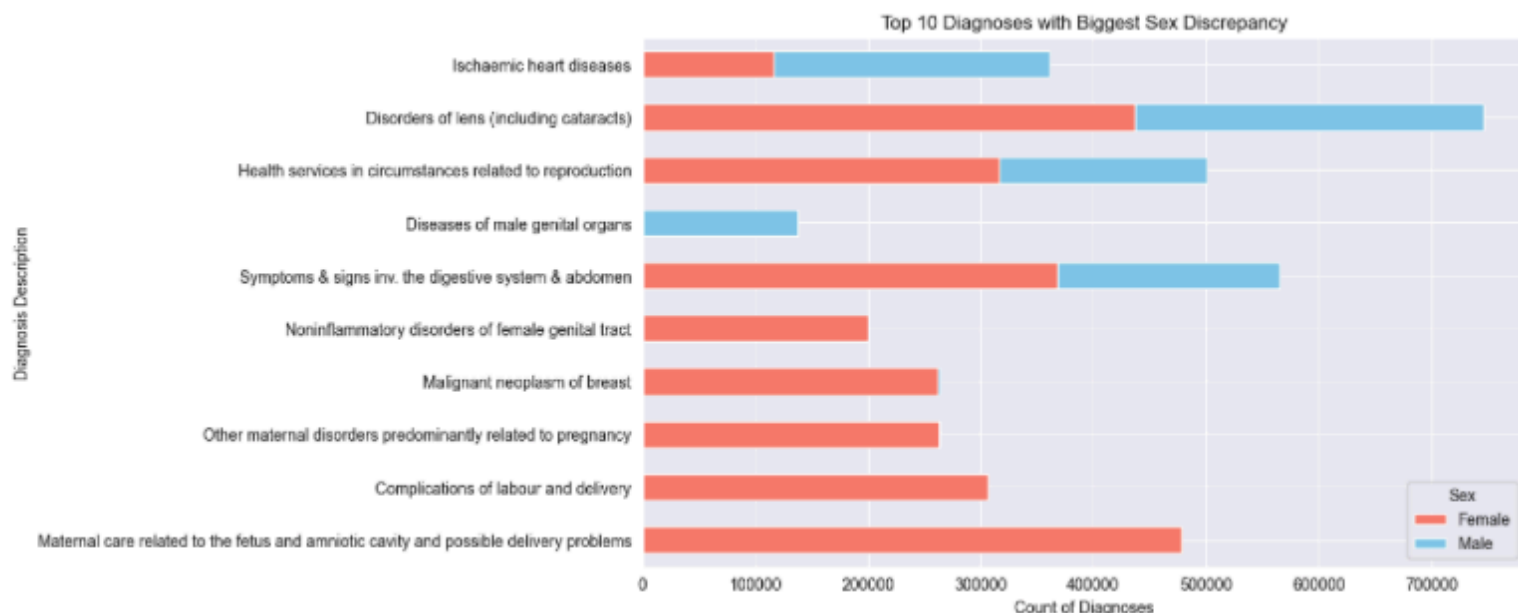
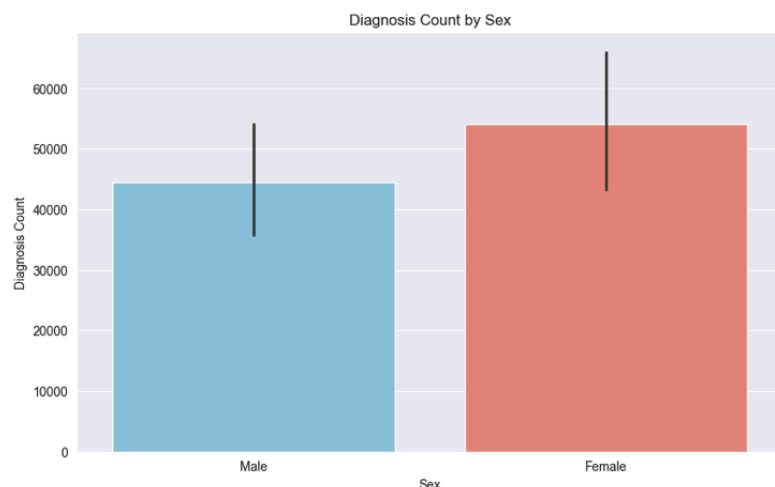
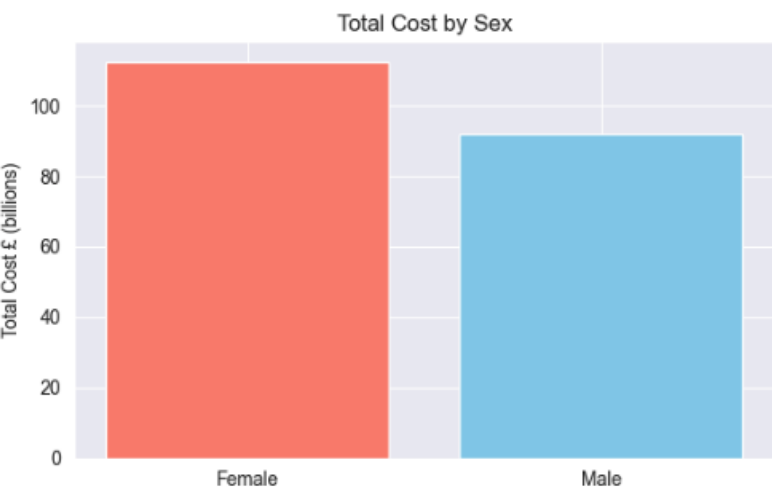
1. UK healthcare data was split by country
 - England has the richest datasets; Scotland and Wales use separate systems.
 - Solution: Focused on NHS England and acknowledged limitations. Supplement with additional datasets, if time allowed.
2. The datasets did not cover identical years
 - The datasets covered different time periods, and so didn't align to the same year
 - Solution: Used the most recent datasets available and focused on trend insights rather than strict cross-year comparisons.
3. Costings in the dataset did not directly separate sexes for all services
 - Some categories were aggregated
 - Solution: Used available age-band + sex combinations; where needed, estimated costs using activity count x average cost.
4. Non-binary and trans identities were not included
 - Most NHS cost datasets are sex - based only
 - Solution: Acknowledged this limitation explicitly in the report
5. Aggregated data meant fewer outliers in healthcare cost data
 - Extreme cases sometimes distorted averages

RESULT REPORTING

Overall, our findings indicated that women contributed a disproportionately higher share of NHS healthcare costs compared to men. This appeared largely driven by sex-specific health needs, particularly those associated with reproductive health, maternity, and age-related conditions such as menopause. These results are consistent with the diagnostic data, which shows significantly higher treatment volumes for gynaecological and maternity-related issues among women.

In contrast, men tend to incur higher costs in diagnostic categories typically associated with male-dominant health risks, such as cardiovascular diseases. Furthermore, our analysis of diagnosis frequency by sex revealed that women seek medical support more often than men, leading to a higher number of recorded diagnoses. This aligned with existing literature suggesting that women are generally more proactive in seeking healthcare intervention, while men may delay medical consultation [2][1]

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CONCLUSION

The evidence appeared to have consistently indicated that:

- Women utilised hospital services more so than men
- Women received higher diagnosis counts in most categories
- The estimated hospital expenditure calculated using sex proportions suggested that women incur a greater total cost. Population size alone did not explain the difference
- Overall, healthcare costs are rising, which may increase the cost disparity

Some suggested improvements to the existing framework that the project produced would include:

- Add predictive modelling for future cost differences
- Apply age-standardisation to remove demographic skew (age distribution can distort the comparison)

Overall findings indicated that women incurred higher healthcare costs than men, which is largely driven by greater hospital activity and higher diagnosis counts. Biological and social factors play a role, but system-level cost trends can also influence these differences.

SUPPORTING REFERENCES

[1] Wang et al., 2013 – “Do men consult less than women?” (BMJ Open)

<https://bmjopen.bmj.com/content/3/8/e003320?>

- Found that men’s consultation rates were about 32% lower than women’s overall, even after adjusting for some health needs.

Supports: “Women use primary care services more; men under-consult / delay seeking care.”

[2] Ballering et al., 2023 – “Sex and gender differences in primary care help-seeking behaviour” (Family Practice) <https://pmc.ncbi.nlm.nih.gov/articles/PMC10193899>

- Showed that biological sex (being female), more than gender identity/expression, is associated with higher help-seeking for physical symptoms.

Supports: The idea that sex itself is a strong predictor of using primary care more often.

[3] [NHS hospital costs by sex and age, NHS Digital, “Acute patient level activity and costing 2018–19: Age and gender”](#)

Supports: Cost burden patterns differ by sex and across the life course.

Contributors (In no particular order)

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